

Homework for Chapter 4

Calculate the spectra of reflection for a single surface for no damping, weak ($\gamma=0.1 \Delta_{LT}/\hbar$) and strong damping ($\gamma=0.6 \Delta_{LT}/\hbar$). Compare the shift of the reflection maxima and minima with respect to the transverse and longitudinal eigenenergies ω_b and ω_l , respectively. Which quantity can be deduced with reasonable accuracy from a first inspection of the reflection spectra?

Below is the optical phonon frequency in GaAs

Phonon Type	Abbreviation	Frequency (THz)	Frequency (cm ⁻¹ , wavenumber)
Transverse Optical	TO	~8.1	~270
Longitudinal Optical	LO	~8.9	~297

The static dielectric constant of GaAs is 12.9

$$\frac{\varepsilon_s}{\varepsilon_b} = \frac{\omega_l^2}{\omega_0^2} > 1 \quad \text{for} \quad f > 0$$

$$\varepsilon_s = \varepsilon_b + \frac{f}{\omega_0^2}$$

$$\varepsilon(\omega) = \left(\varepsilon_b + \frac{f(\omega_0^2 - \omega^2)}{(\omega_0^2 - \omega^2)^2 + \omega^2 \gamma^2} + i \frac{\omega \gamma f}{(\omega_0^2 - \omega^2)^2 + \omega^2 \gamma^2} \right)$$

$$= \varepsilon_1(\omega) + i \varepsilon_2(\omega)$$

$$n(\omega) = \left(\frac{1}{2} \left\{ \varepsilon_1(\omega) + [\varepsilon_1^2(\omega) + \varepsilon_2^2(\omega)]^{1/2} \right\} \right)^{1/2}$$

$$\kappa(\omega) = \left(\frac{1}{2} \left\{ -\varepsilon_1(\omega) + [\varepsilon_1^2(\omega) + \varepsilon_2^2(\omega)]^{1/2} \right\} \right)^{1/2}$$

$$R(\omega) = \frac{I_r}{I_i} = \frac{[n(\omega) - 1]^2 + \kappa^2(\omega)}{[n(\omega) + 1]^2 + \kappa^2(\omega)}$$